

Correction Methods to Leapfrog Numerical Integrator

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Abstract

This paper investigates the challenges of operating numerical integration systems with discontinuous forces, with a focus on the second-order leapfrog integrator as used in astrophysical simulations. Standard leapfrog methods exhibit systematic errors when encountering force discontinuities that lead to inaccurate trajectories and accumulating errors. We aim to correct a critical asymmetric drift issue at the boundary crossing points through two developed correction methods, one that changes the velocity calculations and one that adjusts the timestep size. These methods have shown improvements in error analysis tests against the standard method for discontinuous models through convergence properties. We have also identified limitations regarding smoother force transitions, exhibiting systematic energy drift that compromises the symplectic nature of the original integrator.

All code related to this research, including the implementation of the integrators and test models is available publicly through a GitHub repository: <https://github.com/tj-griffiths/Research>.

Introduction

The current focus of this research is to investigate numerical integration methods for systems with discontinuous forces, with applications to self-interacting dark matter simulations. The method used is a second-order leapfrog integrator that follows three steps:

1. Velocity update: $v_{i+1} = v_i + \frac{dt}{2} \cdot \text{force}(x_i)$
2. Position update: $x_{i+1} = x_i + dt \cdot v_{i+1/2}$
3. Velocity update: $v_{i+1} = v_i + \frac{dt}{2} \cdot \text{force}(x_{i+1})$

This algorithm staggers the updates of position and velocity, giving it the characteristic "leapfrog" name. The symplectic nature of this method ensures that energy is conserved over long integration periods for Hamiltonian systems, making it particularly valuable for astronomical simulations where energy conservation is critical to maintaining correct orbital dynamics.

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Test Models

Using a harmonic oscillator as a test model, we analyzed the energy conservation properties of the method in different implementations and dimensionalities. Further work on the more complex case of a bouncing ball is used as a step in modeling particle collisions. The bouncing ball model utilized two forces, one based on the absolute value of the position and the other using an arctangent function with a scaling factor to allow for direct comparison.

1. Absolute Value Force: A direct implementation of a discontinuous force using the absolute value function: $\text{force}(x) = -g \cdot \text{sign}(x)$.

2. Arctangent Force: A smoothed approximation of the discontinuous force using an arctangent function with a scaling factor: $\text{force}(x) = -g \cdot \frac{2}{\pi} \cdot \arctan(\frac{x}{x_s})$.

Where g is the gravitational constant and the scaling factor x_s controls the sharpness of the transition near $x = 0$, allowing us to investigate how the abruptness of the force discontinuity affects numerical error and stability.

Drift Identification

Through detailed analysis of phase space diagrams and trajectory extrema, we identified a critical issue in the standard leapfrog implementation when applied to systems with discontinuous forces: asymmetric drift near the boundary crossing points (where $x = 0$). This drift manifests itself as unequal steps across the boundary, where $\Delta x_{x<0} \neq \Delta x_{x>0}$, resulting in an antisymmetric distortion of the phase space.

The drift occurs because the kick right before the point of contact is either too large or too small compared to the kick after the contact, causing a systematic bias in the trajectory that accumulates over multiple orbits. This effect becomes particularly significant for larger timesteps or sharper discontinuities in the force law.

Correction Method

To address drift issues, we developed and tested several correction methods that are implemented at the crossing point $x_i \cdot x_{i+1} \leq 0$.

Velocity Correction

Our first approach involved a sophisticated correction based on a detailed analysis of the leapfrog algorithm at the crossing point. We define the crossing time by solving for t in the change in time of the position calculation:

$$x(t) = x_i + v_{i+1/2}(t - t_i) \rightarrow t_c = -\frac{x_i}{v_{i+1/2}} + t_i. \quad (1)$$

Then, plugging t_c into the velocity equations, we obtain:

$$v_b = v_i + \frac{-x_i}{v_{i+1/2}} \cdot \text{force}(x_i) \quad (2)$$

$$v_{i+1} = v_b + \left[\left(\frac{x_i}{v_{i+1/2}} + dt\right) \cdot \text{force}(x_{i+1})\right] \quad (3)$$

Here, v_b is the adjusted velocity step before the crossing. This approach precisely calculates when the crossing occurs between timesteps and appropriately adjusts the velocity kicks to maintain symmetry across the boundary.

Timestep Correction

This approach modifies the integration timesteps near the crossing point:

$$\Delta(t_1) = -x_i/v_{i+1} \quad (4)$$

$$\rightarrow v_{i+1} = v_i + \text{force}(x_i) \cdot \Delta(t_1) \quad (5)$$

$$\Delta(t_2) = dt - \Delta(t_1) \quad (6)$$

$$\rightarrow v_{i+1} = v_i + \text{force}(x_{i+1}) \cdot \Delta(t_2) \quad (7)$$

The velocity and position updates are adjusted using these modified timesteps as seen in (5) and (7). This effectively implements an adaptive timestep method that ensures accurate handling of the boundary crossing while maintaining the overall integration scheme.

Energy Conservation

Energy conservation in numerical integration is fundamental to the accurate simulation of physical systems over long timescales. We conducted a systematic investigation of energy conservation properties for the arctangent force model with varying degrees of discontinuity (controlled by the scale factor x_s) across multiple timestep sizes.

Method and Performance

Our analysis examined the energy evolution for the standard leapfrog integrator and both correction methods on a parameter grid of timestep sizes ($dt = 10^{-4}$ to 10^{-1}) and scale factors ($x_s = 10^{-3}$ to 10^0). Figure 1 presents this comprehensive view of energy conservation behavior.

The energy conservation characteristics vary dramatically on the basis of the sharpness of the force discontinuity. For sharp discontinuities of $x_s = 10^{-3}$, the standard leapfrog method exhibits stable, bounded energy oscillations with minimal long-term drift. Correction methods produce step-like changes that correspond to boundary crossings with irregular fluctuations that increase with timestep size. In the

moderate discontinuous regime ($x_s = 10^{-2}$ to 10^{-1}), the standard method maintains relatively consistent energy levels throughout most time steps. The correction methods exhibit identical behavior at smaller timestep size. In the continuous regime ($x_s = 10^0$), both correction methods exhibit systematic energy drift. For the smallest timestep, the energy decreases monotonically over time.

Energy Drift Mechanisms

For more continuous forces, the precise identification of the boundary crossing becomes ambiguous where the force changes gradually rather than at a precise point. As both correction methods are based on accurate boundary detection, we observe systemic errors when the detection is imprecise. For the velocity correction term, the velocity updates depend on the estimated crossing time, which may become overcompensated for with the extended transition region in the continuous case. The timestep correction method follows a similar systemic bias as a result of these estimated crossing times.

Comparative Error Analysis

We evaluated the three methods with the two force models across a range of stepsizes (10^{-4} to 10^0) to track the trajectories of both the position and velocity. The error analysis was conducted using the absolute values of difference in the final values of both the reference solution (a standard model with a timestep of $1e-8$) and each method. This error metric captures the cumulative the integration errors throughout the simulation timeline, and provides a measure of how well each method maintains the correct trajectory through the endpoints.

Results of Position and Velocity Error (Fig. 2)

In the arctangent model, the standard errors follows the expected stepsize squared relationship, where the vertical separation is dependent on the scale factor ($x_s = 1$, continuous in this test). The convergence to h^2 is derived from the Taylor series expansions of any smooth function (like an arctangent curve) and the relationship between local truncation error and global error. The local truncation error has a leading term that is proportional to h^3 for both the position and velocity. However, when integrating over a fixed time interval we use $n = \text{time}/h$ steps, therefore the global error is $O(h^3) * T/h = O(h^2)$. The correction methods challenge this smoothness and lead to sharp peaks and troughs while still following the overall proportional trend. The correction methods begin to vary as a stepsize of 10^{-2} while still remaining on a similar track. The methods begin to veer off at larger stepsizes where there is a loss of asymptotic behavior where the higher-order terms in the Taylor expansion are non-negligible.

The absolute value showcases a true discontinuous model, where the convergence behavior is lost. In the regime of small timesteps, the correction methods outperform the standard model, maintaining a better adherence to the second-order convergence. In contrast, at larger stepsizes the correction methods exhibit a greater sensitivity to larger timesteps.

Results of Maximum Energy Error (Fig. 3)

A further investigation on the numerical performance of the integration methods was conducted on how the maximum energy error scales with stepsize for both forces (both of which discontinuous with an arctangent scale factor of $x_s = 0.001$). This approach used a maximum relative energy error throughout the simulation, providing a more stringent test that was less dependent on the phase alignment of the solutions.

In the arctangent case (top panel), the standard method shows superior energy conservation at the smallest stepsizes, which disappears for practical stepsizes above 10^{-3} . The opposite is true for a pure discontinuous force, where the correction methods substantially better energy conservation at stepsizes less than 10^{-1} , following similar results to the error analysis of the position and velocity. The regime of higher stepsize becomes nearly identical for both forces, with all methods converging to similar error magnitudes. The loss of the asymptotic behavior in the arctangent force is due to the more discontinuous scale factor, where the higher-order terms dominate even at the smaller stepsizes.

Conclusion

This research has demonstrated the importance of properly addressing discontinuities in numerical integration methods when accurately simulating systems with abrupt force transitions. The developed correction systems have shown improvement with potential applications to proper systems. The limitations to more continuous force regimes as showcased in the energy conservation analysis (Fig. 1, top left panel) where the correction methods exhibited systematic energy loss over time are a challenge to be addressed. This distinction in the behaviors across the two forces highlights how the systems require the optimization in their own respective regimes. Further directions of work look toward a symplectic correction method as it would combine the trajectory accuracy with long term stability that is essential for astronomical simulations.

Figures

Energy Conservation - Arctan Force

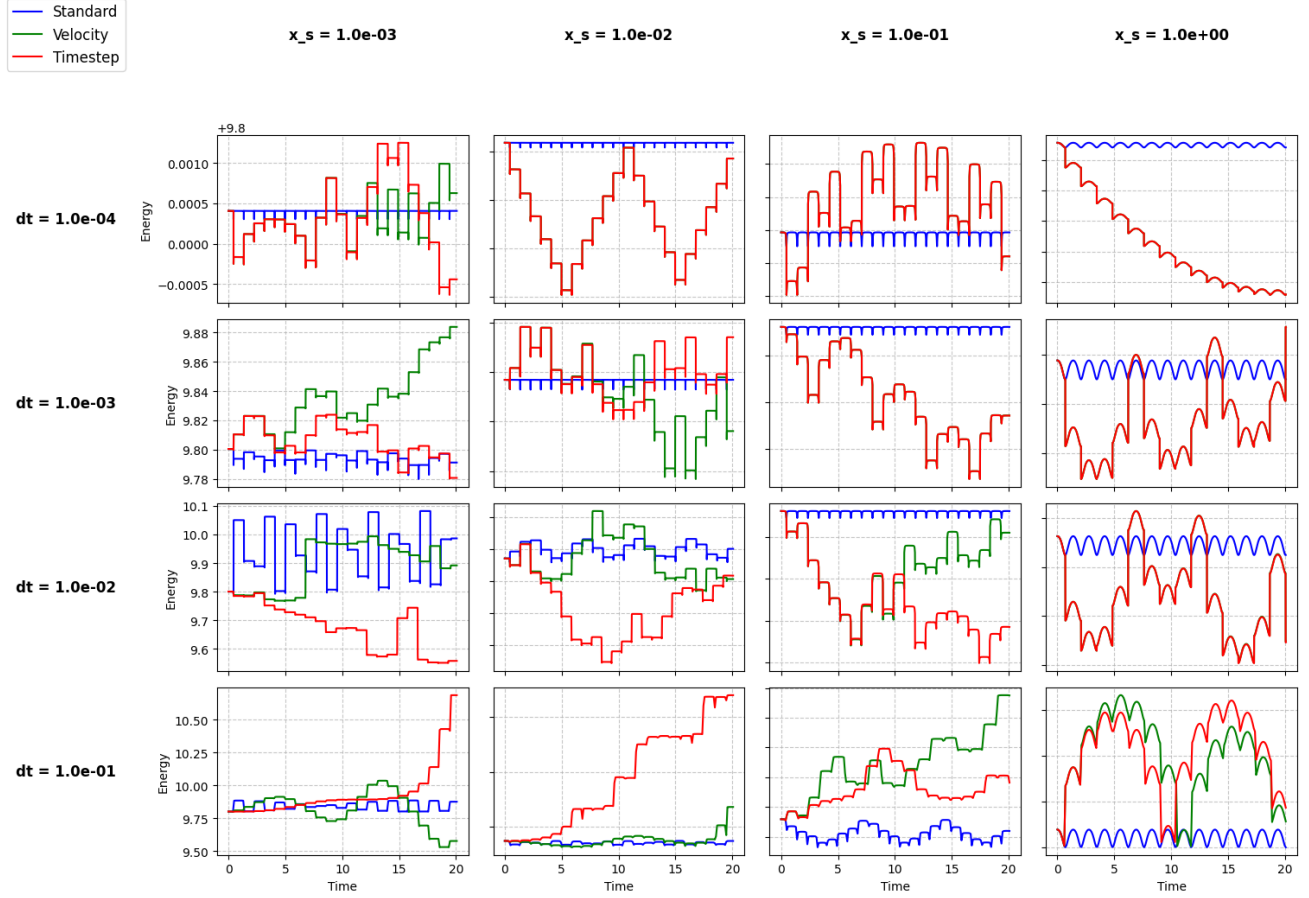


Figure 1: The grid displays the Energy vs Time of the Standard model of leapfrog and the two correction methods (Velocity corrected and Timestep Corrected) with the Arctangent Force. The subplots are organized to show varying timestep and scale factor, where the rows are constant in timestep (indicated by the magnitude on the left) and the columns are constant in scale factor (indicated by the magnitude above). The cases in which the Velocity Method (green line) is not visible, the result follows that of the Timestep Method.

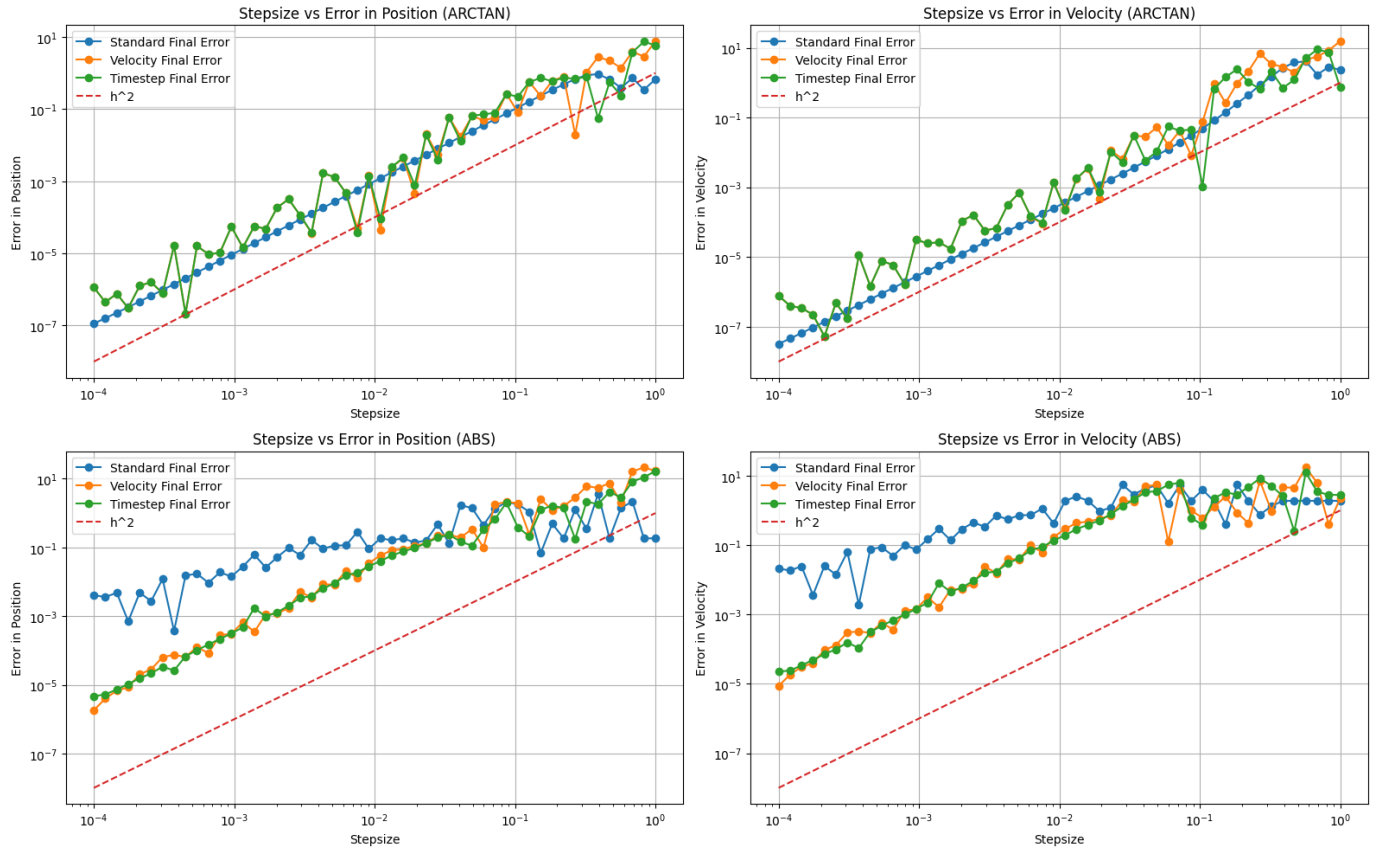


Figure 2: The four plots showcase the Error Analysis in the position and velocity for both the arctangent and absolute Value forces. Included are the three models, the standard solution and two correction methods that are each compared to the expected reference line given by a stepsize (h) squared relationship.

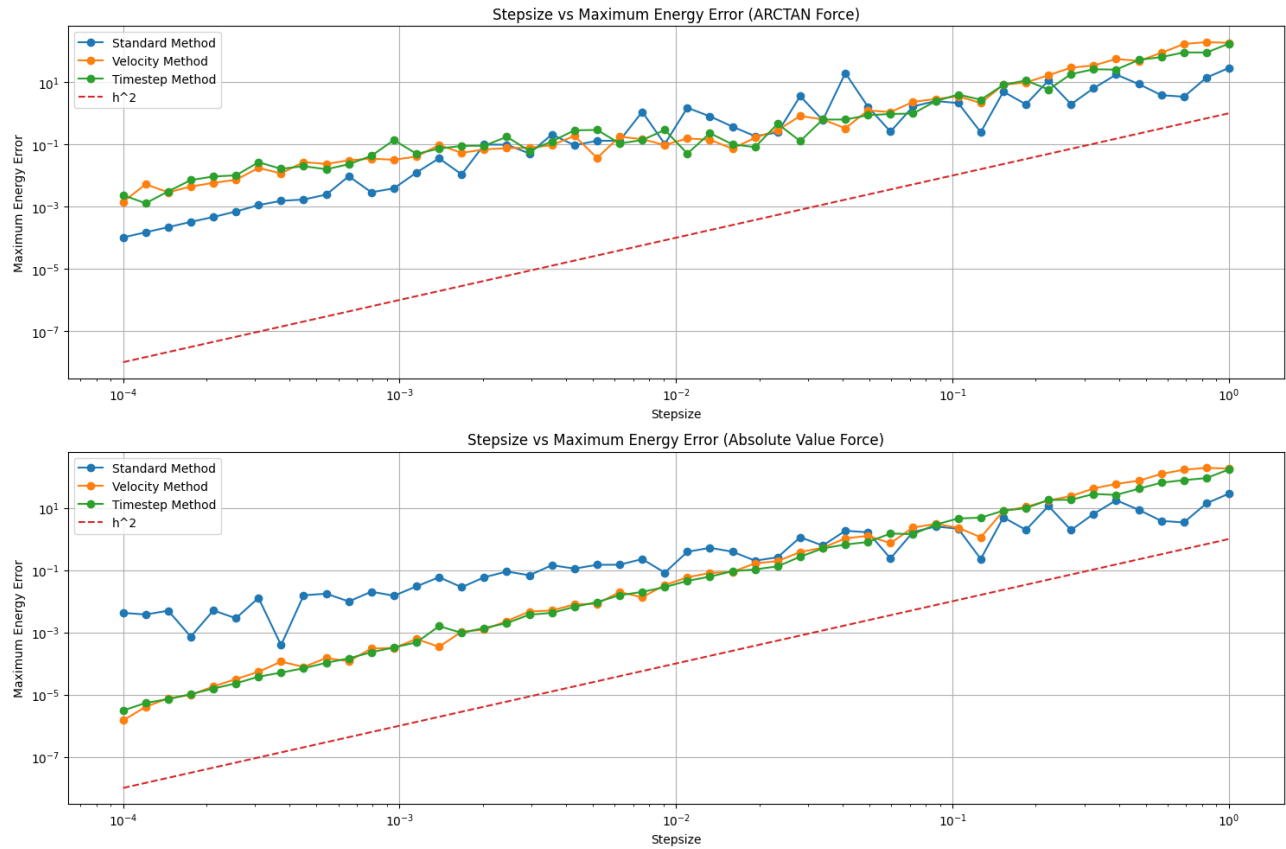


Figure 3: A comparative display of the Stepsize vs Maximum Energy Error across a continuous (ARCTAN) and discontinuous (ABS) model with the three methods. Included is the expected proportional relationship following the stepsize (h) squared.